

Enhancing user prompt confidentiality in Large Language Models through advanced differential encryption

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ABSTRACT

In the era of artificial intelligence (AI) advancements heralded by Large Language Models (LLMs) like GPT-3, the capacity to parse and generate human-like text brings to light substantial privacy concerns. These arise notably from LLMs' reliance on vast datasets often laden with personal information, underscoring the potential for inadvertent memorization and disclosure of sensitive data. Addressing these pivotal privacy concerns, our research introduces a novel two-fold approach aimed at bolstering the confidentiality and security of user data in LLM applications. Firstly, we deploy advanced cryptographic techniques, incorporating bespoke encryption and hashing protocols, to preprocess user data. This strategy effectively anonymizes personal identifiers prior to their processing by LLMs, directly tackling the challenges of sensitive information exposure. Concurrently, our methodology encompasses a secure mutual authentication protocol utilizing lightweight cryptographic measures. This ensures that system interactions are strictly reserved for authenticated users, thereby enhancing overall data security. Collectively, our approach not only preserves the utility of data for AI tasks but also fortifies the privacy framework surrounding LLMs, significantly reducing the likelihood of privacy breaches and steering AI development towards a more secure and ethically grounded future.

1. Introduction

Recent advancements in natural language processing (NLP) and artificial intelligence (AI), notably propelled by the development of advanced large language models (LLMs) like GPT-3 and BERT, have opened new horizons in digital world [1]. These models, built

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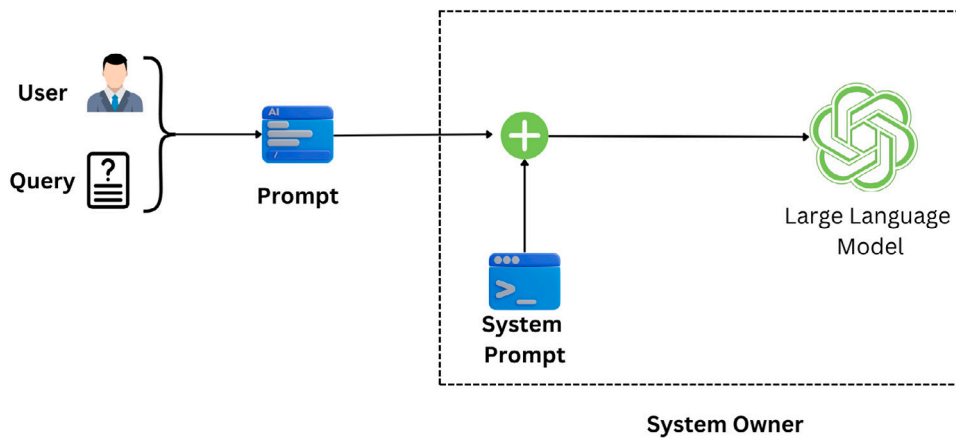


Fig. 1. Working of large language model.

on sophisticated transformer neural networks and deep learning methodologies, have not only excelled in traditional NLP benchmarks such as sentiment analysis, machine translation, and speech recognition [2] but also demonstrated capabilities in assimilating common sense knowledge and handling complex ethical scenarios, which are crucial in dynamic urban environments [3,4].

As we integrate these technologies into smart devices, significant concerns around data privacy and security emerge. The reliance of LLMs on extensive language datasets, including those focused on phonemics essential for refining language modeling and speech recognition techniques [5], necessitates a stringent approach to data protection and ethical usage. This is especially critical given their application in areas where fine-tuning for specific tasks, including smart transport systems, requires careful consideration of both the utility and the ethical implications of the data being processed [6–8].

The ethical dimensions of deploying these models in public domains. Addressing inherent political biases and ensuring ethical development and deployment is essential for the responsible integration of AI solutions [9,10].

Within the operational schema of large language models (LLMs), the interaction sequence commences with the user, who articulates a query necessitating the model's response (Fig. 1). This query is then translated into an AI-compatible prompt, which reformulates the user's initial request into structured input tailored for the LLM's comprehension. After this translation, the system may introduce its augmentations to the prompt, potentially enhancing the query with additional data or refining the prompt to better align with the LLM's operational paradigms.

The refined prompt is then relayed to the LLM, the technological cornerstone of the system, which employs its extensive training on diverse textual corpora to generate a pertinent response. The system owner, delineating the periphery of this interactive network, oversees the LLM's performance and ensures the seamless execution of the model's functionalities. Moreover, the system owner bears the onus of safeguarding the privacy and security of the data coursing through the LLM, a responsibility of heightened significance in the contemporary digital milieu. Fig. 1 encapsulates this intricate exchange, illustrating not only the user's engagement with the AI through prompts but also the system owner's pivotal role in the maintenance and governance of the LLM.

In large language models (LLMs), the privacy implications of using prompts have become a critical concern in recent scholarly discourse. A pivotal study by Mahendran et al. [11], Tembhurne et al. [12], Bisht and Vampugani [13] delved into the privacy vulnerabilities inherent in contemporary language models, mainly focusing on the risks of pattern reconstruction and keyword inference attacks. This research accentuates the urgency of addressing privacy issues within natural language processing as LLMs find broader applications in various fields.

Additionally, the research trajectory in prompt learning has increasingly gravitated towards utilizing large-scale language models through prompts [14–16]. While prompts are crucial in guiding these models to execute specific tasks, they also bring forth privacy challenges, particularly when sensitive data might be unintentionally disclosed or misused. Recent studies have shed light on the privacy hazards linked to language models, underscoring the imperative for ongoing research into privacy-preserving text generation methods and the establishment of metrics to evaluate privacy maintenance during text generation [17,18]. These concerns highlight the necessity for robust privacy-protective strategies to counter the risks posed by prompts in LLMs.

Furthermore, exploring prompt-based techniques for controllable text generation has revealed the benefits and flexibility of using natural language prompts to steer models towards specific tasks or topics [19,20]. While these methods offer significant control in text generation, they also bring privacy concerns to the fore, especially when dealing with sensitive or confidential data. The potential privacy ramifications of these prompt-based generation methods call for meticulous attention and the development of advanced techniques to protect sensitive information.

The assessment of privacy risks in text data remains a dynamic area of research, emphasizing the necessity to evaluate the privacy implications of language models, particularly in their interaction with sensitive data [21,22]. As LLMs continue to advance and find new applications, it is imperative to thoroughly assess and mitigate the privacy risks associated with prompts to safeguard data privacy.

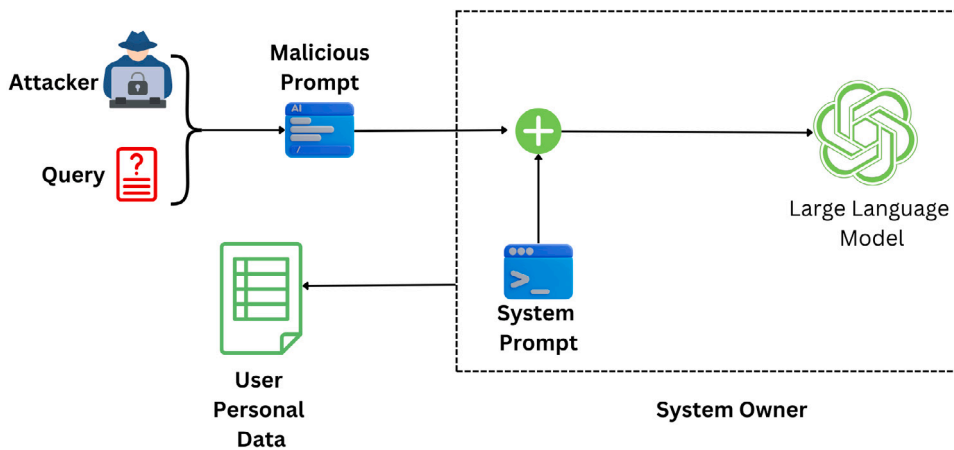


Fig. 2. Attack scenario.

In this context, this study makes several significant contributions to the field of privacy and security in Large Language Models (LLMs). Firstly, we introduce an advanced cryptographic approach tailored for LLMs, enhancing the confidentiality of user prompts without compromising the model's performance. Secondly, our work proposes a secure mutual authentication protocol, ensuring that only authorized users can interact with the system, thereby protecting against unauthorized access and potential data breaches. Lastly, through rigorous theoretical analysis and empirical validation, our research not only addresses the immediate privacy concerns associated with LLMs but also lays a foundation for future developments in secure AI interactions. These contributions collectively advance the state of the art in secure and ethical AI operations, establishing a new benchmark for privacy-centric approaches in the era of advanced language models.

The Rest of the paper is organized as follows: Motivation is presented in Section 2. Section 3 presents the recent developments in the field of LLMs. Our proposed approach is presented in Section 4. Section 5 presents the security proof of our proposed approach. Lastly, Section 6 concludes the paper.

2. Motivation

Fig. 2 represents the common attack scenario that the attackers use. According to Fig. 2 an adversary seeks to extract personal data from a Large Language Model (LLM). This is achieved by submitting crafted queries to elicit responses containing sensitive user information. The process begins with the attacker formulating a query that may seem benign but is intended to manipulate the LLM into disclosing private data. This query is inputted as a 'Malicious Prompt' into the system.

The system, designed to interface with the LLM, includes a 'System Prompt' mechanism as a safeguard. This component assesses the incoming prompts for potential threats or privacy violations. If a prompt is identified as malicious, the system prompt can alter or nullify it to prevent the LLM from processing it further. However, should the malicious prompt bypass this defensive layer, it reaches the LLM. Since LLMs are trained on extensive corpora that could contain user-specific data, there is a risk that the model could inadvertently output information that compromises user privacy.

Upon receiving a sanitized or unsanitized prompt, the LLM generates a response based on its training. The attacker aims to analyze this response to extract valuable personal data. The dashed boundary encapsulates the system owner's domain, indicating control over the LLM and the System Prompt. The system owner is responsible for implementing robust security measures to prevent such attacks and safeguard user data.

3. Related work

In the rapidly evolving field of large language models, a comparative analysis of several prominent models highlights their diverse architectures, parameter counts, specializations, and unique features. As detailed in Table 1, the Flan-T5-large, based on the T5 architecture, is known for its large parameter count and general NLP specialization, with its unique text-to-text approach making it highly versatile for various tasks. Dolly-6B, characterized by a GPT-like architecture and a 6 billion parameter count, is specifically geared towards creative content generation. LLaMA-13B and Vicuna-13B, both with a GPT-like architecture and 13 billion parameters, are effective in general NLP tasks, with Vicuna-13B focusing on language understanding and specific task adaptations. The Cerebras-GPT-13B, similar in size to the latter models, stands out for its optimization with Cerebras Systems' hardware.

GPT-NEOX-20B and Flan-UL2 (20B), each with a 20 billion parameter count, are large models with a general NLP focus; the former extends the GPT architecture, while the latter's hybrid design offers improved adaptability and learning efficiency. Vicuna-13B-v1.3, an updated version of Vicuna-13B, inherits its predecessor's specialization with additional improvements and

Table 1
Comparative analysis of large language models.

Model	Architecture	Parameter Count	Specialization	Unique Features
Flan-T5-large	T5-based	Large	General NLP	Text-to-text approach, versatile with prompts
Dolly-6B	GPT-like	6 Billion	Creative content	Geared towards artistic content generation
LLaMA-13B	GPT-like	13 Billion	General NLP	Effective in a range of NLP tasks
Vicuna-13B	GPT-like	13 Billion	Language understanding	Adaptations for specific language tasks
Cerebras-GPT-13B	GPT-based	13 Billion	General NLP	Optimized for Cerebras Systems' hardware
GPT-NEOX-20B	GPT-extension	20 Billion	General NLP	Large size for complex language tasks
Flan-UL2 (20B)	Hybrid	20 Billion	General NLP	Hybrid model for improved adaptability and learning efficiency
Vicuna-13B-v1.3	Updated Vicuna-13B	13 Billion	Language understanding	Improvements and optimizations over original model
Llama2-13B-chat	Conversational AI	13 Billion	Conversational AI	Fine-tuned for chatbot applications
ChatGPT-4	GPT-optimized	4th Generation	Conversational AI	Specialized in interactive and responsive communication

optimizations. Llama2-13B-chat and ChatGPT-4, both with a conversational AI specialization, are tailored for chatbot applications and interactive communication. Llama2-13B-chat is fine-tuned for chatbot applications, while ChatGPT-4, optimized for the GPT architecture, excels in responsive communication? This comparative analysis, summarized in Table 1, provides insights into these advanced language models' varied capabilities and applications.

The security paradigms of large language models (LLMs) have gained pronounced attention within the research community, catalyzing the creation of multifaceted frameworks to bolster ethical, privacy, and safety measures. Within cloud computing, a market-inspired framework has been posited to fortify self-adaptive security protocols within inherently dynamic cloud environments [23]. This particular framework advocates for a fluid approach to security that can evolve in tandem with the mutable threat landscape characteristic of cloud computing.

Additionally, the advancement of an adaptive authentication and authorization model has been tailored specifically for the intricacies of service-oriented enterprise computing. This model proffers an intelligent security engine capable of bridging the security voids present in conventional Identity and Access Management (IDAM) frameworks, thus offering a more robust approach to authentication and authorization for service consumption [24]. Such a model underlines the necessity for security frameworks that can adapt to the distinct requirements of service-oriented architectures. Such a model underlines the necessity for security frameworks that can adapt to the distinct requirements of service-oriented architectures. Xu et al. [25] proposed a public auditing scheme with data privacy. Liu et al. [26] proposed security for IoT devices. Zhou et al. [27] proposed Gan-siamese network for user privacy. Zhang et al. [28] presents user security for cyber-physical social services.

In healthcare, a novel framework dedicated to generating synthetic training data has been crafted to de-identify electronic health records, with a keen focus on safeguarding privacy through automated text generation. This framework also introduces metrics to appraise the efficacy of privacy preservation in text generation, marking a significant stride in protecting sensitive health data [17].

The conversation around the ethical dimensions of LLMs has also intensified, particularly concerning the moral agency these models simulate. The development of ethical frameworks to navigate intelligent systems' behavior, including LLMs, has become a pivotal focus, recognizing the imperative for moral guidance in AI system implementation [29].

Concurrently, in scholarly publishing, the rise of LLMs like GPT-3 has instigated ethical and security deliberations, pondering the prospective impacts on research integrity and publishing practices. This discussion extends into the design of pediatric mHealth interventions for adolescents with complex social needs, where ethical frameworks are being proposed to ensure that privacy and ethical considerations are paramount [30].

4. Proposed approach

In our proposed approach, we introduce a two-phase process to enhance the security and privacy in the utilization of Large Language Models (LLMs).

4.1. Securing the prompts

Incorporating the proposed cryptographic model for enhancing privacy in the utilization of LLMs, as shown in Fig. 3, we formalize the process where multiple users $U_1, U_2, \dots, U_n$ contribute datasets D_i containing personal information. The objective is to apply a transformation T that effectively obfuscates sensitive information within these datasets. This transformation process can be mathematically articulated as follows i;

$$D'_i = T(D_i), \quad (1)$$

Here, D'_i represents the anonymized or encrypted dataset corresponding to user 'i', with the personal information either removed or securely encrypted.

Within the transformation (T), cryptographic measures are utilized. These can include encryption (E) using an essential K, followed by a hashing function (H), yielding a compressed and secure representation of the data;

$$D'_i = H(E(D_i, K)), \quad (2)$$

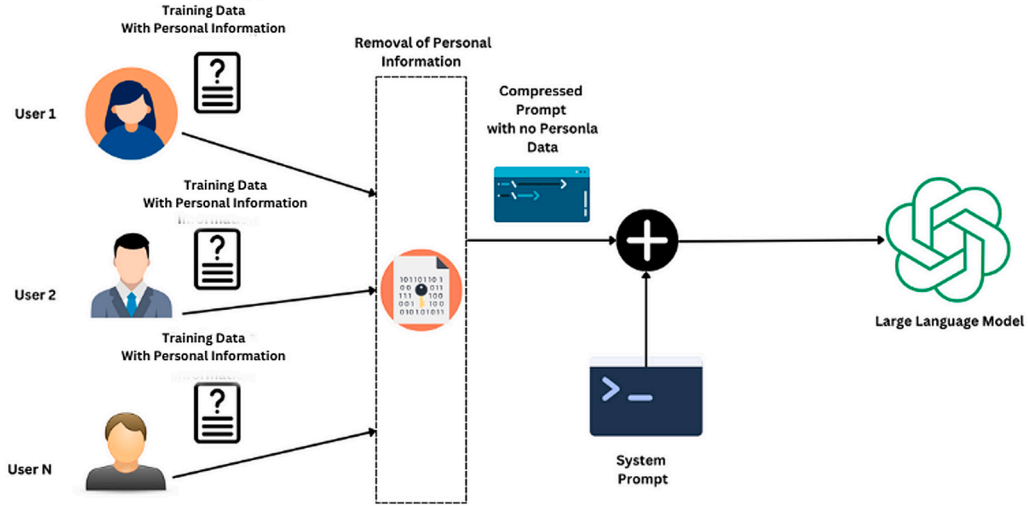


Fig. 3. Proposed framework.

Where $E(D_i, K)$ encrypts the data, making it accessible only to entities possessing the essential K , H further processes this encrypted data into a fixed-length format, resistant to reverse-engineering attempts.

The transformed datasets from all users are then combined into a single, sanitized dataset D' ;

$$D' = \bigcup_{i=1}^n D'_i, \quad (3)$$

From this sanitized dataset, a system prompt P is generated, which is devoid of any personal data and is suitable for processing by the LLM:

$$P = \text{GeneratePrompt}(D') \quad (4)$$

Finally, the LLM accepts the prompt P and computes a response O ;

$$O = \text{LLM}(P). \quad (5)$$

The entire process, as outlined in Fig. 3, ensures that at no point does the LLM directly interact with or process any personal information. The outputs (O) generated by the LLM are based solely on the sanitized data, thus maintaining the privacy of the individual users' data.

4.2. Secure authentication of users

The second phase is dedicated to providing a secure authentication phase for users utilizing a Lightweight Cryptographic (LC) technique. This phase is designed to verify the identity of users before they can submit prompts to the system, adding an additional layer of security. By leveraging LC techniques, the system can efficiently authenticate users without imposing significant computational overhead, thus maintaining a seamless user experience while ensuring robust security measures are in place.

4.2.1. User registration phase

In this phase, user (U_a) registers himself with the cloud operator (Co) to obtain the credentials that must be authenticated among the user-server to access the serverless service environment.

- **Credential Generation:** User's smart device such as a handset or AR devices is considered as his identification, i.e. ID. After that, the user selects a unique password (PW) and his biometric (Bio) conditions in the devices. As the biometric information keeps on changing, in our proposed model, we used a fuzzy system to generate the steady biometric variable (σ);

$$(\sigma, \delta) \leftarrow \text{Gen}(\text{Bio}_i), \quad (6)$$

Where $\text{Gen}()$ is the fuzzy system. After that, the system selects a random nonce $l \in \mathbb{Z}^*$; this nuance protects the user ID and password.

- After selecting the identification values, user calculates Tid and Hp_w from the following equations;

$$Tid \leftarrow h(ID \parallel l), \quad (7)$$

$$Hpw \leftarrow l \oplus h(PW \parallel \sigma), \quad (8)$$

where h is the hash function.

- The cloud server selected a public key(K_u) and a random number (p_i). After that the cloud provider calculates Hid , A_i , and C_i from the following equations;

$$Hid \leftarrow h(Tid \parallel K_u), \quad (9)$$

$$A_i \leftarrow h(Hpw \parallel Tid) \oplus Hid, \quad (10)$$

$$C_i \leftarrow h(Tid \parallel Hpw) \oplus p_i, \quad (11)$$

After calculating Hid , A_i , and C_i the cloud operator send these values to the user;

$$U_a \xleftarrow{A_i, C_i, h(.)} Co, \quad (12)$$

- After reviving $A_i, C_i, h(.)$ from the cloud operator (Co), the smart device calculates R_1 and R_2 from the following equation;

$$R_1 \leftarrow h(Pw \parallel l) \parallel (A_i \oplus \sigma), \quad (13)$$

$$R_2 \leftarrow h(ID \parallel R_1 \parallel l), \quad (14)$$

After calculation of R_1 and R_2 the device stores following values in the memory;

$$\{R_1, R_2, Hpw, A_i, C_i\} \xleftarrow{Stores} Device,$$

4.3. Third party registration phase

In this phase, the side unit (RoC)(M_a) registers itself with the cloud operator (Co) to obtain the credentials that must be authenticated among the user-server to access the serverless environment.

- Credential Generation: M_a 's selected his identification, i.e. ID. After that, the user selects a unique secret key (M_{PW}). After that, the M_a selects a random nonce $m \in \mathbb{Z}^*$.
- After M_a calculates $MTid$ and $MHpw$ from the following equations;

$$MTid \leftarrow h(ID \parallel l), \quad (15)$$

$$MHpw \leftarrow l \oplus h(M_{PW}), \quad (16)$$

where h is the hash function. After that, M_a transfers these values to the cloud operator.

- The cloud service provider selected a public key(K_u) and a random number (p_j). After that the cloud provider calculates $MHid$, MA_j , and MC_j from the following equations;

$$MHid \leftarrow h(MTid \parallel K_u), \quad (17)$$

$$MA_j \leftarrow h(MHpw \parallel MTid) \oplus MHid, \quad (18)$$

$$MC_j \leftarrow h(Tid \parallel Hpw) \oplus p_i, \quad (19)$$

After calculating Hid , MA_j , and MC_j the cloud operator send these values to the user.

$$M_a \xleftarrow{MA_j, MC_j, h(.)} Co \quad (20)$$

- After reviving $MA_j, MC_j, h(.)$ from the cloud operator (Co), the M_a calculates MR_1 and MR_2 from the following equation;

$$MR_1 \leftarrow h(M_{PW} \parallel l) \parallel (MA_j \oplus \sigma), \quad (21)$$

$$MR_2 \leftarrow h(MID \parallel MR_1 \parallel l), \quad (22)$$

After calculation of MR_1 and MR_2 the device stores following values in the memory;

$$\{MR_1, MR_2, Hpw, MA_j, MC_j\} \xleftarrow{Stores} Device,$$

5. User login phase

When a user wants to access the serverless services, he enters his ID^* , biometrics (Bio^*), and password (Pw^*), the AR device. Then the AR device checks the ID^* and password (Pw^*) using following method;

$$\sigma^* \leftarrow Gen(Bio^*), \quad (23)$$

$$l^* \leftarrow Hpw \oplus h(Pw^* \parallel \sigma^*), \quad (24)$$

$$R_1^* \leftarrow h(Pw^* \parallel i^*) \parallel (A_i \oplus \sigma^*), \quad (25)$$

$$R_2^* \leftarrow h(ID^* \parallel R_1^* \parallel l^*), \quad (26)$$

If $R_2 = R_2^*$ login successful. Therefore, in this phase, only authenticated users get logged in to the device.

5.1. Authentication phase

To access the RoC provider's services, a user has to connect with the RoC service provider through the cloud operator. So, firstly, the user selects a time stamp (T_1); this timestamp is used to avoid the replay attack. After the selection of the timestamp, the user calculates;

$$W_1 \leftarrow h(Tid \parallel Pid), \quad (27)$$

$$J_1 \leftarrow X_i d \oplus W_1, \quad (28)$$

$$J_2 \leftarrow h(Pid \parallel Tid \parallel T_1), \quad (29)$$

$$Co \xleftarrow{\{J_1, J_2, Pid, T_1\}} User, \quad (30)$$

After receiving, $\{J_1, J_2, Pid, T_1\}$ from the user first checks the time stamp and then the user exists or not through the user's Pid value. Then the cloud operator calculates J_2^* as follows;

$$if \ T_2 - T_1 \leq \Delta T \text{ noreplay attack}, \quad (31)$$

$$J_2^* \leftarrow h(Tid^* \parallel T_1 \parallel Pid), \quad (32)$$

$$if \ J_2^* = J_2 \text{ userauthenticated}, \quad (33)$$

After the authentication of the user, the cloud operator extracts the value of Service provider server that the user wants to connect through following equations;

$$X_j \leftarrow J_1 \oplus h(Tid \parallel pid), \quad (34)$$

X_j is the service provider server that the user wants to connect. The cloud service provider stores this value in the user table for the record. After this, the cloud operator sends this information to the service provider through the following equations;

$$MW_1 \leftarrow h(MTid \parallel MPid), \quad (35)$$

$$MJ_1 \leftarrow U_i d \oplus MW_1, \quad (36)$$

$$MJ_2 \leftarrow h(MPid \parallel MTid \parallel T_2), \quad (37)$$

$$Service \ Provider \xleftarrow{\{MJ_1, MJ_2, MPid, T_2\}} Co, \quad (38)$$

After receiving $\{MJ_1, MJ_2, MPid, T_2\}$ from the cloud operator first checks the time stamp, Then the service operator calculates MJ_2^* as follows;

$$if \ T_3 - T_2 \leq \Delta T \text{ noreplay attack}, \quad (39)$$

$$MJ_2^* \leftarrow h(MTid^* \parallel T_2 \parallel MPid), \quad (40)$$

$$if \ MJ_2^* = MJ_2 \text{ userauthenticated}, \quad (41)$$

After the authentication of the user, the cloud operator extracts the value of service provider server that the user wants to connect through following equations;

$$User_p \leftarrow MJ_1 \oplus h(MTid \parallel Mpid). \quad (42)$$

After this phase, the service provider gets the active users' ID, and the active communication starts.

6. Results and discussion

6.1. BAN logic analysis

BAN logic, developed by Burrows, Abadi, and Needham, is a formal method used to verify the security of cryptographic protocols [31]. This logic is particularly effective in proving the correctness and security of cryptographic authentication protocols [32]. It is widely recognized as a valuable tool for modeling and verifying authentication in cryptographic protocols [33]. By assuming perfect cryptographic functions, BAN logic can systematically identify any incorrect designs in authentication protocols [34].

Researchers have successfully utilized BAN logic to prove mutual authentication in various protocols [35]. It is a well-known formal logic technique used for analyzing the security of cryptographic protocols [36]. BAN logic has been extensively employed to

Table 2
BAN logic symbols.

Notation	Meaning
$P \models X$	Principal P believes X
$P \triangleleft X$	Principal P sees (or receives) X
$P \diamond X$	Principal P once said X
$P \subset X$	Principal P controls the truth of X
$X \#$	X is fresh (has not been sent in a message before)
$\{X\}_K$	X is encrypted with essential K
$P \leftrightarrow KQ$	Principals P and Q share key K
$P \Rightarrow K$	K is the public key of P
$P \Leftarrow K$	K is the secret (private) key of P
$\#(X)$	The nonce (number used once) value of X

provide mutual authentication proof in cryptographic protocols across different works [37]. Moreover, it has been used to formally verify authentication schemes in remote healthcare systems [38].

The application of BAN logic extends to proving correctness in mutual authentication features of protocols [39]. It has been instrumental in validating security services in NFC mobile payment systems [40]. Additionally, BAN logic has been applied to analyze the security of cryptographic protocols, ensuring mutual authentication and key agreement [36]. It is a logic framework widely accepted for assessing authentication protocols [32]. The explanations of the BAN logic symbols are shown in Table 2.

6.1.1. Logical postulates

Numerous restrictions exist to check session key sharing in BAN logic. The guidelines provided by BAN logic are as follows;

- Message Meaning Rule

$$P \models (Q \leftrightarrow K) \wedge P \triangleleft \{X\}_K \Rightarrow P \models (Q \diamond X),$$

- Nonce Verification Rule

$$P \models (Q \diamond X) \wedge P \models (\#(X)) \Rightarrow P \models (Q \models X),$$

- Jurisdiction Rule

$$P \models (Q \subset X) \wedge P \models (Q \models X) \Rightarrow P \models X,$$

- Freshness Conjunction Rule

$$P \models (\#(X)) \wedge P \models (\#(Y)) \Rightarrow P \models (\#(X, Y)),$$

- Seeing Conjunction Rule

$$P \triangleleft (X, Y) \Rightarrow P \triangleleft X \wedge P \triangleleft Y,$$

- Belief in Public Keys

$$P \models (Q \Rightarrow K) \wedge P \triangleleft \{X\}_{K^{-1}} \Rightarrow P \models (Q \diamond X),$$

- Belief in Shared Secrets

$$P \models (P \leftrightarrow KQ) \Rightarrow P \models (Q \models (P \leftrightarrow KQ)),$$

- Belief in Freshness

$$P \models (\#(X)) \wedge P \models (P \diamond X) \Rightarrow P \models (\#(P \diamond X)),$$

6.1.2. Security goals

: These goals help to check the success of mutual authentication;

- Goal of Mutual Belief in Shared Secret Key

$$P \models (P \leftrightarrow KQ) \wedge Q \models (P \leftrightarrow KQ),$$

- Goal of Message Origin Authentication

$$P \models (Q \diamond X),$$

- Goal of Message Content Acceptance

$$P \models X,$$

- Goal of Freshness Assurance

$$P \models (\#(X)),$$

- Goal of Secure Key Distribution

$$P \models (Q \Rightarrow K),$$

- Goal of Mutual Authentication

$$P \models (Q \models (P \diamond X)),$$

6.1.3. Assumptions

We used following assumptions in our proposed model

- Initial Belief in Shared Secret Key

$$P \models (P \leftrightarrow KQ),$$

- Initial Belief in Public Key Authenticity

$$P \models (Q \Rightarrow K),$$

- Initial Belief in the Control over Proposition

$$P \models (Q C X),$$

- Initial Belief in the Freshness of Nonces

$$P \models \#(N),$$

- Initial Belief in the Integrity of Messages

$$P \models (Q \models X),$$

- Initial Belief in Possession of Credentials

$$P \models (P \text{ has } X),$$

- Initial Belief in the Security of Encryption

$$P \models (\{X\}_K \text{ is secure}),$$

BAN logic proof

- P_{St} receives $Msg1$:

$$G1 : P_{St} \triangleleft \{avta, Aa, U1, T1\}_{U2}.$$

- Applying MMR utilizing $G1$ and $A1$, $G2$ is obtained:

$$G2 : P_{St} \models Ua \sim (avta, Aa, U1, T1).$$

- Applying FR utilizing $G2$ and $A2$, $G3$ is obtained:

$$G3 : P_{St} \models \#(avta, Aa, U1, T1).$$

- Integrating $G2, G3$ along with NVR application yields :

$$G4 : P_{St} \models Ua \models (avta, Aa, U1, T1).$$

- Applying BR on $G4$, we obtain $G5$:

$$G5 : P_{St} \models Ua \models (avta, Aa, U1, T1),$$

- Utilizing $G5$, P_{St} computes ;

$$W2 = h(Aa \parallel avta \parallel r2 \parallel xt \parallel T2),$$

$$U2 = xt.U1 \text{ and } U4 = w2.Pa.$$

Thus P_{St} can compute the session key as:

$$SK = h(avta \parallel U2 \parallel U4 \parallel T2),$$

which further results in the generation of $G6$,

$$G6 : P_{St} \models Ua \models (Ua \leftrightarrow P_{St}) \text{ (Goal-4),}$$

- Applying JR utilizing $G6$ and $A5$, $G7$ is obtained.

Table 3
Comparative analysis.

Model	Login and authentication cost (in bits)
[41]	1632
[42]	1824
[43]	2080
[44]	1440
[45]	1536
Proposed	1320

$G7 : P_{S_i} \models (Ua \leftrightarrow P_{S_i})$ (Goal-3).

- Ua receives $Msg2$:

$G8 : Ua \triangleleft \{avta, Aa, U3, T2\}_{U4}$.

- Applying MMR utilizing $G8$ and $A3$, $G9$ is obtained:

$G9 : Ua \models P_{S_i} \sim (avta, Aa, U3, T2)$.

- Applying FR utilizing $G9$ and $A4$, $G10$ is obtained:

$G10 : Ua \models \#(avta, Aa, U3, T2)$.

- Integrating $G9, G10$ along with NVR application yields :

$G11 : Ua \models P_{S_i} \models (avta, Aa, U3, T2)$.

- Applying BR on $G11$, we obtain $G12$:

$G12 : Ua \models P_{S_i} \models (avta, Aa, U3, T2)$.

- Utilizing $G12$, Ua computes:

$W1 = h(Aa \parallel avta \parallel r1 \parallel xa \parallel T1)$,

$U2 = w1.P_{K_i}$ and $U4 = xa.U3$.

Thus Ua can successfully generate:

$SK = h(avta \parallel U2 \parallel U4 \parallel T2)$,

which results in the generation of $G13$,

$G13 : Ua \models P_{S_i} \models (Ua \leftrightarrow P_{S_i})$ (Goal-2).

- Applying JR utilizing $G13$ and $A6$, $G14$ is obtained:

$G14 : Ua \models (Ua \leftrightarrow P_{S_i})$ (Goal-1).

6.2. Comparison analysis

In this section, we compare our proposed models with other available models (see Table 3).

In cybersecurity and digital authentication, efficiency is often quantified regarding computational and communication overheads. One critical metric used to evaluate the effectiveness of an authentication system is the login and authentication cost, which is measured in bits. This metric essentially represents the amount of data transmitted during the authentication process, which is a crucial determinant of the system's speed and resource consumption. Our analysis focuses on a comparative study of the proposed authentication model against several established models in the field. We included models developed by Panda and Chattopadhyay [41], Wang et al. [42], Li et al. [43], Ryu et al. [44], and Son et al. [45]. The comparison pivots on the Login and Authentication Cost, offering a clear view of how our proposed model stacks up against its contemporaries.

In determining the computational costs associated with our model, we meticulously quantified the bit lengths of various security elements and operations. Specifically, the random nonces R_1 , R_2 , and R_3 each require 160 bits, mirroring the bit length allocated for the device identity ID_i and each of the security parameters $\{CK, T_i, V_i\}$. Furthermore, the output of the hash function, specifically SHA-1 in this context, also adheres to a 160-bit length, aligning with the standard output size for this cryptographic hash function. Notably, our security model leverages the efficiency of Elliptic Curve Cryptography (ECC), where the combined point $P = (P_x, P_y)$ on the curve requires a total of 320 bits. This allocation is based on the understanding that the security strength of a 160-bit ECC key is commensurate with that of a 1024-bit RSA cryptosystem, offering a robust security framework while optimizing for computational efficiency. This detailed account of bit allocation underscores the rigorous approach to ensuring data security within our model, while also highlighting the model's alignment with established cryptographic standards.

Starting with the model by Panda and Chattopadhyay [41], which exhibits a Login and Authentication Cost of 1632 bits, it is evident that while their model is efficient, there is room for improvement in terms of data overhead. Haq et al.'s approach marks a slightly higher cost at 1824 bits, suggesting a more data-intensive process, potentially offering higher security at the cost of

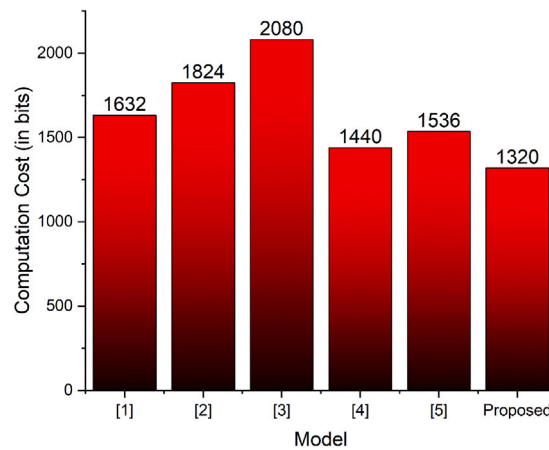


Fig. 4. Computation cost comparison.

efficiency. Li et al.'s model further escalates this cost to 2080 bits, which, while potentially providing robust security features, also implies more significant resource usage during authentication.

In contrast, Ryu et al. [44] and Son et al. [45] have developed models that lean towards greater efficiency. Ryu et al.'s model costs 1440 bits, and Son et al.'s model comes close to 1536 bits. These figures balance security and efficiency, aligning closely with contemporary needs for rapid and secure application authentication processes. However, our proposed model sets a new benchmark in this domain with a Login and Authentication Cost of merely 1320 bits. This reduced cost signifies not only an improvement in terms of efficiency but also reflects the optimization of communication overheads without compromising the integrity and security of the authentication process. This efficiency is crucial in environments with limited bandwidth or applications where rapid authentication is paramount (see Fig. 4).

7. Conclusion

Incorporating advanced cryptographic techniques into Large Language Models (LLMs) signifies a major leap forward, marrying cutting-edge AI capabilities with the imperative of rigorous data protection. This research delineates a two-fold approach that showcases the potential of employing LLMs across a wide array of applications while staunchly safeguarding user privacy. The first facet of our strategy involves the application of tailored encryption and hashing protocols before data is processed by the LLM framework, establishing a robust privacy safeguard that maintains the integrity and functionality of the models. Concurrently, the second aspect introduces a secure mutual authentication protocol, reinforcing the system's defense by ensuring that only authorized users can interact with the LLM, thereby enhancing the overall security posture. This dual strategy underscores our commitment to responsible AI development, where the cultivation of user trust is deemed as critical as technological innovation. Our approach not only sets a precedent for the integration of secure AI in complex, real-world scenarios but also lays the groundwork for future research and deployment of privacy-centric AI solutions. By addressing current privacy concerns and advocating for a future where AI technologies are developed with a keen eye on ethical implications, this study contributes to steering the AI domain towards a more secure, privacy-aware, and ethically responsible trajectory.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- [1] Trummer I. From bert to gpt-3 codex. *Proc VLDB Endow* 2022;15:3770–3. <http://dx.doi.org/10.14778/3554821.3554896>.
- [2] Yang J, Wang M, Zhou H, Zhao C, Zhang W, Yu Y, Li L. Towards making the most of bert in neural machine translation. In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 34, 2020, p. 9378–85. <http://dx.doi.org/10.1609/aaai.v34i05.6479>.
- [3] Kovalev A, Panov A. Application of pretrained large language models in embodied artificial intelligence. *Dokl Math* 2022;106:S85–90. <http://dx.doi.org/10.1134/s1064562422060138>.
- [4] Hu B, Gaurav A, Choi C, Almomani A. Evaluation and comparative analysis of semantic web-based strategies for enhancing educational system development. *Int J Semant Web Inf Syst (IJSWIS)* 2022;18(1):1–14.
- [5] Kłosowski P. A rule-based grapheme-to-phoneme conversion system. *Appl Sci* 2022;12:2758. <http://dx.doi.org/10.3390/app12052758>.
- [6] Trummer I. Database tuning using natural language processing. *ACM Sigmod Rec* 2021;50:27–8. <http://dx.doi.org/10.1145/3503780.3503788>.
- [7] Almomani A, Alauthman M, Shatnawi MT, Alweshah M, Alrosan A, Alomoush W, et al. Phishing website detection with semantic features based on machine learning classifiers: a comparative study. *Int J Semant Web Inf Syst (IJSWIS)* 2022;18(1):1–24.
- [8] Pan X, Yamaguchi S, Kageyama T, Kamlin MHB. Machine-learning-based white-hat worm launcher in botnet defense system. *Int J Softw Sci Comput Intell (IJSSCI)* 2022;14(1):1–14.
- [9] Liu R, Jia C, Wei J, Xu G, Wang L, Vosoughi S. Mitigating political bias in language models through reinforced calibration. In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 35, 2021, p. 14857–66. <http://dx.doi.org/10.1609/aaai.v35i17.17744>.
- [10] Madhu S, Padunnavallappil S, Saajlal PP, Vasudevan VA, Mathew J. Powering up an IoT-enabled smart home: A solar powered smart inverter for sustainable development. *Int J Softw Sci Comput Intell (IJSSCI)* 2022;14(1):1–21.
- [11] Mahendran D, Luo C, McInnes B. Review: privacy-preservation in the context of natural language processing. *IEEE Access* 2021;9:147600–12. <http://dx.doi.org/10.1109/access.2021.3124163>.
- [12] Tembhurne JV, Almin MM, Diwan T. Mc-DNN: Fake news detection using multi-channel deep neural networks. *Int J Semant Web Inf Syst (IJSWIS)* 2022;18(1):1–20.
- [13] Bisht J, Vampugani VS. Load and cost-aware min-min workflow scheduling algorithm for heterogeneous resources in fog, cloud, and edge scenarios. *Int J Cloud Appl Comput (IJCAC)* 2022;12(1):1–20.
- [14] Li D, Ye S. Using bidirectional prompt learning in nlp few shot tasks. *Front Comput Intell Syst* 2023;3:167–72. <http://dx.doi.org/10.54097/fcis.v3i1.6362>.
- [15] Li S, Qin D, Wu X, Li J, Li B, Han W. False alert detection based on deep learning and machine learning. *Int J Semant Web Inf Syst (IJSWIS)* 2022;18(1):1–21.
- [16] Kumbhojkar NR, Menon AB. Integrated predictive experience management framework (IPEMF) for improving customer experience: In the era of digital transformation. *Int J Cloud Appl Comput (IJCAC)* 2022;12(1):1–13.
- [17] Libbi C, Trienes J, Trieschnigg D, Seifert C. Generating synthetic training data for supervised de-identification of electronic health records. *Future Internet* 2021;13:136. <http://dx.doi.org/10.3390/fi13050136>.
- [18] Gupta BB, Gaurav A, Arya V. Navigating the security landscape of large language models in enterprise information systems. *Enterprise Inf Syst* 2024;2310846.
- [19] Schick T, Udupa S, Schütze H. Self-diagnosis and self-debiasing: a proposal for reducing corpus-based bias in nlp. *Trans Assoc Comput Linguist* 2021;9:1408–24. http://dx.doi.org/10.1162/tacl_a_00434.
- [20] Nhi NTU, Le TM, et al. A model of semantic-based image retrieval using C-tree and neighbor graph. *Int J Semant Web Inf Syst (IJSWIS)* 2022;18(1):1–23.
- [21] Xiong P, Lin L, Zhu Y, Zhu T. Pritxt: a privacy risk assessment method for text data based on semantic correlation learning. *Concurr Comput Pract Exp* 2021;34. <http://dx.doi.org/10.1002/cpe.6680>.
- [22] Gaurav A, Psannis K, Peraković D. Security of cloud-based medical internet of things (miots): A survey. *Int J Softw Sci Comput Intell (IJSSCI)* 2022;14(1):1–16.
- [23] Tziakouris G, Mera-Gómez C, Ramírez F, Bahsoon R, Buyya R. Market-inspired framework for securing assets in cloud computing environments. *Softw Pract Exp* 2022;52:2005–25. <http://dx.doi.org/10.1002/spe.3113>.
- [24] Ibrahim M, Mohamed B, Hassan M. An adaptive authentication and authorization model for service oriented enterprise computing. *Kuwait J Sci* 2021;49. <http://dx.doi.org/10.48129/kjs.v49i1.10745>.
- [25] Xu Z, He D, Vijayakumar P, Gupta BB, Shen J. Certificateless public auditing scheme with data privacy and dynamics in group user model of cloud-assisted medical WSNs. *IEEE J Biomed Health Inf* 2021;27(5):2334–44.
- [26] Liu RW, Guo Y, Lu Y, Chui KT, Gupta BB. Deep network-enabled haze visibility enhancement for visual IoT-driven intelligent transportation systems. *IEEE Trans Ind Inf* 2022;19(2):1581–91.
- [27] Zhou Z, Li Y, Li J, Yu K, Kou G, Wang M, et al. Gan-siamese network for cross-domain vehicle re-identification in intelligent transport systems. *IEEE Trans Netw Sci Eng* 2022.
- [28] Zhang Q, Guo Z, Zhu Y, Vijayakumar P, Castiglione A, Gupta BB. A deep learning-based fast fake news detection model for cyber-physical social services. *Pattern Recognit Lett* 2023;168:31–8.
- [29] Hagendorff T, Danks D. Ethical and methodological challenges in building morally informed ai systems. *AI Ethics* 2022;3:553–66. <http://dx.doi.org/10.1007/s43681-022-00188-y>.
- [30] Bounds D, Stiles-Shields C, Schueller S, Odgers C, Karnik N. Ethical considerations for developing pediatric mhealth interventions for teens with socially complex needs. *J Child Adolescent Psychiatric Nurs* 2022;36:7–16. <http://dx.doi.org/10.1111/jcap.12396>.
- [31] Kawamoto Y. An epistemic approach to the formal specification of statistical machine learning. *Softw Syst Model* 2020;20:293–310. <http://dx.doi.org/10.1007/s10270-020-00825-2>.
- [32] Jung J, Moon J, Lee D, Won D. Efficient and security enhanced anonymous authentication with key agreement scheme in wireless sensor networks. *Sensors* 2017;17:644. <http://dx.doi.org/10.3390/s17030644>.
- [33] Aloufi B, Alhakami W. A lightweight authentication mac protocol for cr-wsns. *Sensors* 2023;23:2015. <http://dx.doi.org/10.3390/s23042015>.
- [34] Chow M. A secure blockchain-based authentication and key agreement scheme for 3gpp 5g networks. *Sensors* 2022;22:4525. <http://dx.doi.org/10.3390/s22124525>.
- [35] Thammarat C, Kurutach W. A secure fair exchange for sms-based mobile payment protocols based on symmetric encryption algorithms with formal verification. *Wirel Commun Mob Comput* 2018;2018:1–21. <http://dx.doi.org/10.1155/2018/6953160>.
- [36] Jung J, Kang D, Lee D, Won D. An improved and secure anonymous biometric-based user authentication with key agreement scheme for the integrated epr information system. *PLoS One* 2017;12:e0169414. <http://dx.doi.org/10.1371/journal.pone.0169414>.
- [37] Le T, Lu C, Hsu C, Do T, Chou Y, Wei W. A novel three-factor authentication protocol for multiple service providers in 6g-aided intelligent healthcare systems. *IEEE Access* 2022;10:28975–90. <http://dx.doi.org/10.1109/access.2022.3158756>.
- [38] Alanazi M, Nashwan S. Secure and anonymous three-factor authentication scheme for remote healthcare systems. *Comput Syst Sci Eng* 2022;42:703–25. <http://dx.doi.org/10.32604/csse.2022.022962>.
- [39] Mridha M, Imran A, Wadud M, Hamid A. An improved user anonymous secure authentication protocol for healthcare system using wireless medical sensor network. *Int J Comput Digit Syst* 2022;11:251–63. <http://dx.doi.org/10.12785/ijcds/110121>.

- [40] Al-Shammari M, Nashwan S. Fully authentication services scheme for NFC mobile payment systems. *Intell Autom Soft Comput* 2022;32:401–28. <http://dx.doi.org/10.32604/iasc.2022.022065>.
- [41] Panda PK, Chattopadhyay S. A secure mutual authentication protocol for IoT environment. *J Reliab Intell Environ* 2020;6:79–94.
- [42] Wang J, Zhu Y, et al. Secure two-factor lightweight authentication protocol using self-certified public key cryptography for multi-server 5G networks. *J Netw Comput Appl* 2020;161:102660.
- [43] Li Y, Xu M, Xu G. Blockchain-based mutual authentication protocol without CA. *J Supercomput* 2022;78(15):17261–83.
- [44] Ryu J, Son S, Lee J, Park Y, Park Y. Design of secure mutual authentication scheme for metaverse environments using blockchain. *IEEE Access* 2022;10:98944–58.
- [45] Son S, Lee J, Kim M, Yu S, Das AK, Park Y. Design of secure authentication protocol for cloud-assisted telecare medical information system using blockchain. *IEEE Access* 2020;8:192177–91.